



Review article

Bone mineral density in patients with primary ovarian insufficiency: A systematic review and Meta-Analysis

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ABSTRACT

Introduction: Premature menopause is a major complication of primary ovarian insufficiency (POI), and this loss is closely related to bone mineral density (BMD). Previous research has indicated potential associations between BMD and POI. This study set out to provide the first systematic literature review and meta-analysis account of BMD content among women with POI.

Methods: Studies including women with POI and controls were eligible from PubMed, Embase, Cochrane Library and Web of Science databases (from their inception to April 2022). Two reviewers independently evaluated study eligibility. The meta-analysis was performed using the DerSimonian and Laird random effects model.

Results: Ten studies featuring 578 women with POI and 480 controls were selected. BMD content of femur neck (SMD: -0.76; 95 % CI: -1.20 to -0.31; $P = 0.0008$), the BMD content of nondominating forearm (SMD: -0.67; 95 % CI: -1.15 to -0.18; $P = 0.007$) were significantly decreased in women with POI. However, no differences were seen in other regions (lumbar spine, total hip, hipneck).

Discussion: The results of this study indicate that BMD content altered in patients with primary ovarian insufficiency. An implication of this is the possibility that hormone replacement therapy to minimize the prevalence of fracture morbidity and mortality associated with osteopenia in patients with POI.

Introduction

Osteoporosis in postmenopausal women usually occurs 5 ~ 10 years after menopause and affects millions of postmenopausal women worldwide [1]. Osteoporosis is defined as typical progressive systemic skeletal disorder, is characterized by low bone mass, deterioration of bone microstructure, reduction of bone mineral density (BMD) and bone strength, which substantially increases the risk of fracture [2]. The occurrence of osteoporosis is a multifactorial process, and BMD is considered as one of the important indicators for clinical assessment of the severity of osteoporosis [3]. Peak bone mass delays the onset of osteoporosis by 13 years for every 10 % increase, while osteoporosis and fracture is more prevalent in postmenopausal women due to the

deficiency of estradiol, which leads to the enhancement of osteoclast activity and the rapid loss of bone minerals [4].

Young women with hypogonadism are also at increased risk of reduced BMD, and many studies have explored the impact of earlier age at menopause on osteoporosis outcomes [5–7]. Women with premature menopause have 1.5 ~ 3.0 times higher risk of fracture than 49 ~ 55-year-old women who are experiencing menopause [8]. Primary ovarian insufficiency (POI) is defined as women under the age of forty with amenorrhea for at least four months and with a high level of follicle stimulating hormone (FSH) > 25 IU/L (two times checked at least four weeks apart) [9]. There is evidence that premature menopause and chronic estradiol deficiency contributes to osteoporosis, meanwhile, young women with POI have significantly reduced BMD compared with

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women with regular menstruation [10].

Public health benefit can be achieved through the valuable knowledge that BMD may be associated with POI in women in order to reduce their risk of osteoporosis and fractures. POI may provide a better understanding to disentangle the effects of age at menopause on bone loss. For those reasons, the purpose of this systematic review and meta-analysis was to explore whether the content of BMD differed between women with POI and healthy controls.

Methods

Reporting guidelines

This systematic review and meta-analysis was based and reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement [11], registered with the PROSPERO database (<http://www.crd.york.ac.uk/PROSPERO>) under registration number: CRD42020149341. The study followed the guidelines of the Meta-analysis of Observational Studies in Epidemiology (MOOSE)

[12].

Search strategy

PICO (population/intervention/comparison/outcome) included the following components: P (women before the age of 40), I (women with primary ovarian insufficiency), C (healthy women), O (the content of BMD). In order to identify eligible studies, PubMed, Embase, Cochrane Library, and Web of Science databases were searched from their inception to April 2022 (with English as the language; without location or journal restrictions) to identify published studies containing the following keywords: “primary ovarian insufficiency” OR “premature ovarian insufficiency” OR “premature ovarian failure” OR “premature menopause” OR “POI” OR “POF” OR “menopause premature” OR “early menopause” AND “bone density” OR “bone mineral density” OR “BMD” OR “bone” OR “bone mass” OR “bone status” OR “bone structure” OR “bone turnover” OR “bone metabolism” OR “bone mineral content” OR “skeleton” OR “osteoporoses” OR “osteoporosis” OR “osteopenia”.

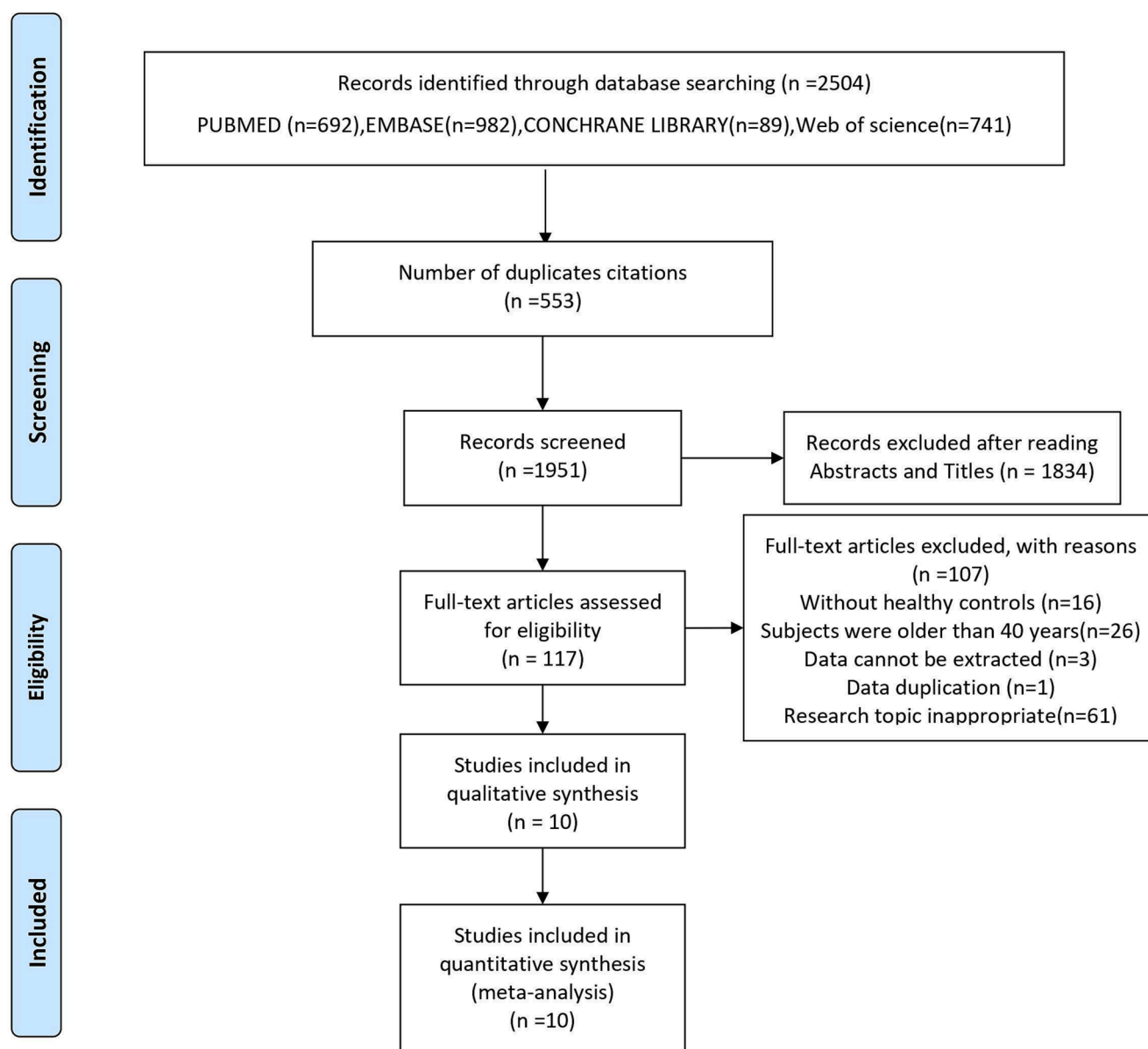


Fig. 1. Flow of the selection process and identification. From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1.

Inclusion and exclusion criteria

The inclusion criteria for eligible studies were as follows: (1) original human observational studies; (2) patients without undergone therapy or gynecological surgery that could trigger menopause; (3) studies reporting the relationship of BMD and POI; and (4) with data on BMD of POI patients and healthy women of similar ages.

Below were the exclusion criteria: (1) non-clinical or animals studies; (2) reviews or case reports; (3) repeated publications; (4) control groups without healthy participants; (5) studies lack of exact data on the content of BMD; (6) studies involved subjects were older than 40 years; and (7) studies with limited sample size (less than 10).

Study selection and data extraction

Two researchers (MJ and YG) independently screened literature and extracted data. A third reviewer was consulted for disagreements between two reviewers (LH). The first step was to import references into EndNote, and to identify duplicates. Following that, abstracts and titles of published literature were screened and excluded based on inclusion criteria. As a final step, the full text of references was reviewed, and the studies were then excluded according to the inclusion and exclusion criteria. As seen in Fig. 1, a flowchart explaining the PRISMA selection process is presented. A complete set of data, including the content of BMD [mean $\pm$ standard deviation (SD)], was extracted from the references, and all the data were double-checked by LH. Each author was contacted and asked for the raw data when continuous variables were absent in reports. In instances where contact failed, standard deviation values were imputed from the median, quartile or ranges, if the data was only displayed graphically, the values were estimated using digital ruler software (Getdata Graph Digitizer, version 2.25).

Quality assessment

Based on the Newcastle-Ottawa Scale (NOS) star system, selected studies were assessed for quality [13], the foregoing studies were included with rating above 7.

Statistical analysis

Meta-analysis was used to analyze the extracted data from the included studies. Due to the inconsistencies in existing references concerning units, the pooled standardized mean difference (SMD) and 95 % confidence intervals (CI) were calculated to evaluate the associations between BMD and POI. Cochran's Q two-sided test was used to test homogeneity in the studies. Heterogeneity in the studies was tested using Cochran's Q two-sided test of homogeneity [14]. In the case of $I^2 < 50\%$, a Mantel-Haenszel fixed-effect model was used. In the event of $I^2 \geq 50\%$, a DerSimonian-Laird random-effects model was used [15,16]. In an attempt to explain heterogeneity, we performed to identify associations between the content of BMD and the characteristics of studies by conducting a subgroup analysis. Meanwhile, in an attempt to test the robustness of the pooled SMD, we performed sensitivity analysis by excluding individual study. The publication bias was measured on the method of funnel plot (in cases where the number of included references were ≥ 10) was used to test publication bias. All analyses were conducted by using RevMan software, and $P < 0.05$ was considered to be statistically significant.

Results

Literature search

Among the 1951 articles, 1834 were excluded after reading titles and abstracts because they did not meet the inclusion criteria. A hundred and seventeen articles were included for full-text assessment, from which

107 were excluded: sixteen without healthy controls; twenty-six involved subjects were older than 40 years; three involved data cannot be extracted; one repeated publication; sixty-one research topic inappropriate. In total, 10 eligible papers (Fig. 1) were included in our analysis, all of which emphasized observational design and used individual data from 578 cases and 480 healthy controls [7,17–25]. The baseline characteristics, such as author, year, region, study design, number of case/control subjects, age, duration of amenorrhea, body mass index (BMI), drug usage, measuring method of BMD and primary conclusion included in the studies could be found in Table 1.

Meta-Analysis results

Meta-analysis of BMD of lumbar spine

Nine studies ($n = 987$ participants) were included in the meta-analysis of BMD content of lumbar spine (Fig. 2A), and there was significant heterogeneity among the studies ($I^2 = 87\%$; $P < 0.00001$). There was no significant association between POI and BMD content of lumbar spine (SMD: -0.32 ; 95 % CI: -0.74 to 0.10 ; $P = 0.14$).

Meta-analysis of BMD of femur neck

Four studies ($n = 521$ participants) compared the BMD content of femur neck between POI women and healthy controls (Fig. 2B), and there was significant heterogeneity among the studies ($I^2 = 80\%$; $P = 0.002$). POI was significantly associated with a decreased BMD content of femur neck (SMD: -0.76 ; 95 % CI: -1.20 to -0.31 ; $P = 0.0008$).

Meta-analysis of BMD of Ward's triangle

Two studies ($n = 354$ participants) compared the BMD content of Ward's triangle between POI women and healthy controls (Fig. 2C). Park's study [19] reported that the BMD content of Ward's triangle in POI group was lower than that in the control group (SMD: -1.14 ; 95 % CI: -1.41 to -0.86 ; $P < 0.00001$). While Leite-Silva's study [24] showed that the BMD content of Ward's triangle in POI group was higher than that in the control group (SMD: 1.10 ; 95 % CI: 0.68 to 1.52 ; $P < 0.00001$).

Meta-analysis of BMD of trochanter

Two studies ($n = 354$ participants) reported the data related to the effects of POI on BMD content of trochanter (Fig. 2D). Park's study [19] revealed that the BMD content of trochanter in POI group was lower than that in the control group (SMD: -0.70 ; 95 % CI: -0.96 to -0.43 ; $P < 0.00001$). While Leite-Silva's [24] study showed that the BMD content of trochanter in POI group was higher than that in the control group (SMD: 1.23 ; 95 % CI: 0.80 to 1.66 ; $P < 0.00001$).

Meta-analysis of BMD of nondominating forearm

Metka's study [17] disclosed that the BMD content of nondominating forearm in POI group was lower than that in the control group (SMD: -0.67 ; 95 % CI: -1.15 to -0.18 ; $P = 0.007$, Fig. 2E).

Meta-analysis of BMD of hip

Mann's study [23] revealed that there was no significant association between POI and the BMD content of total hip (SMD: -0.08 ; 95 % CI: -0.79 to 0.63 ; $P = 0.82$, Fig. 2F), as well as the BMD content of hipneck (SMD: -0.15 ; 95 % CI: -0.85 to 0.56 ; $P = 0.68$, Fig. 2G).

Results of subgroup analysis

There was significant heterogeneity among the studies including in the meta-analysis of serum AMH level of precocious puberty, to search for the sources of heterogeneity and more accurately assess the differences between girls with precocious puberty and healthy controls, subgroup analyses were conducted by geographical location, number of cases, chronological age of cases, the difference between bone age and chronological age of cases, detection reagent (Table 2).

Table 1
Summary characteristics of studies and participants.

No.	Study	Region	Study design	Sample size (n) (Cases vs. Control)		Age(y) (Cases vs. Control)		Duration (y)	BMI(kg/m ²) (Cases vs. Control)		HRT/OC usage		Bone metabolism agentsusage in POI	Measuring method of BMD	Primary conclusion
				POI	Control	POI	Control		POI	Control	POI	Control			
1.	Metka (1992)	Austria (Vienna)	case-control study	41	30	27.12 ± 7.02	23.20 ± 3.34	7.34 ± 9.79	NA	NA	No	No	No	SPA	BMD content of nondominating forearmdecreased in POI
2.	Kurabayashi (1993)	Japan (Niigata)	case-control study	5	35	35.4 ± 2.6	32.6 ± 4.9	NA	22.19	20.50	No	NA	No	DXA- Hologic QDR1000 DPA	BMD content of lumbarvertebra decreased in POI
3.	Park (1995)	Korea (Seoul)	case-control study	91	163	30.2 ± 3.6	NA	3.6 ± 3.6	20.4 ± 3.0	NA	No	No	No		BMD content of femur neck, Ward's triangle, trochanter, and lumbar spinedecreased in POI
4.	Conway (1996)	United Kingdom (London)	retrospective cohort study	122	57	28 (17 to 39)	NA	NA	NA	NA	intermittent use	intermittent use	No	DXA- Hologic QDR1000	BMD content of lumbar spinedecreased in POI
5.	Hartmann (1997)	Austria (Vienna)	case-control study	33	33	28.7 ± 4.9	28.3 + 4.9	2.5 ± 1.3	22.5 ± 3.6	21.6 ± 2.6	No	No	NA	DXA- Hologic QDR1000	BMD content of lumbar spinedecreased in POI
6.	Uygur (2005)	Turkey (Ankara)	case-control study	45	61	33 (18 to 39)	35 (20 to 39)	NA	24.3 ± 4.5	23.5 ± 4.2	intermittent use	No	Thirty-six of the patients received calcium supplementation	DXA	BMD content of femur neck and lumbar spinedecreased in POI
7.	Mann (2008)	UK (London)	case-control study	13	19	35.7 ± 3.4	32.9 ± 5.8	NA	25.2 ± 5.9	24.5 ± 3.7	No	NA	No	DXA-Lunar Prodigy	BMDcontent of spine, hip (total) and hip (neck)remained unchanged in POI
8.	Leite-Silva (2009)	Brazil (Campinas)	case-control study	50	50	29.2 ± 5.9	29.9 ± 6.3	4 ± 3.5	23.9 ± 3.7	25 ± 4.7	Thirty-five of the patients were HRT userswith a mean duration of 28 months of use	NA	No	DXA-Lunar DPX	BMD content of femur neck decreasedin POI; BMD content of lumbar spine, trochanter, and Ward 's triangle increased in POI
9.	Alici- Davutoglu (2013)	Turkey (Istanbul)	case-control study	46	15	28.24 ± 7.85	30.67 ± 4.69	NA	NA	NA	No	No	No	DXA-GE Medical Systems Lunar	BMD content of femur and lumbar spine decreasedin POI
10.	Podfigurna (2020)	Poland (Poznan)	case-control study	132	17	31.86 ± 7.75	23.21 ± 5.86	NA	23.68 ± 4.42	20.71 ± 5.15	No	NA	NA	DXA-Lunar Prodigy Primo	BMD content of lumbar spine decreasedin POI

POI-primary ovarian insufficiency; BMI-body mass index;HRT-hormone replacement therapy; OC-oral contraceptive;BMD-bone mineral density; NA-not available; SPA- single photon absorptiometry; DEX-dual energy X-ray absorptiometry; DPA-dual photon absorptiometry.

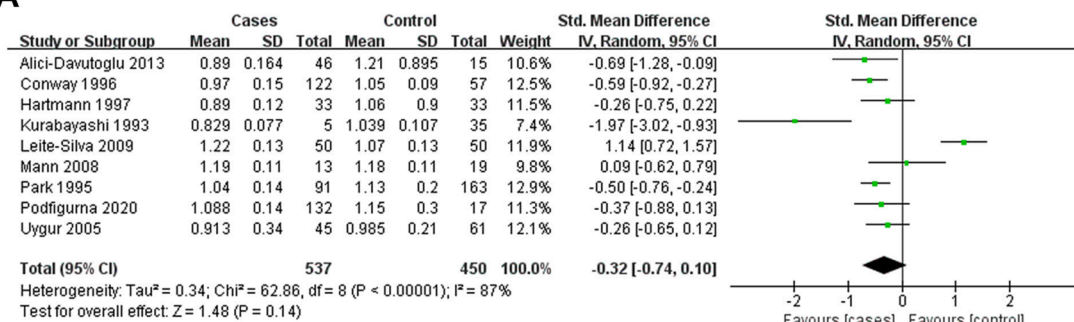
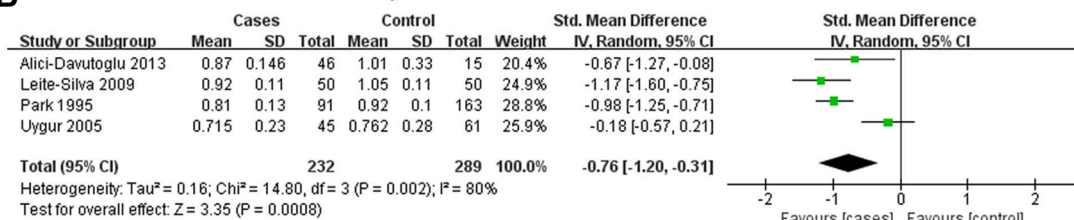
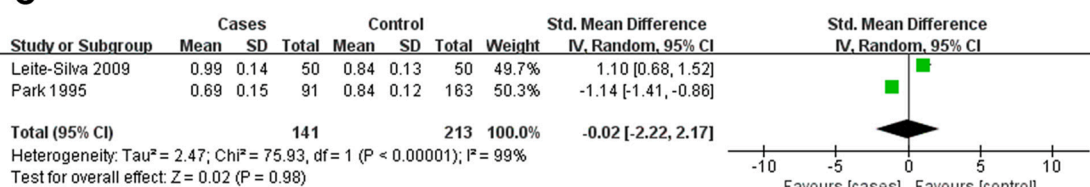
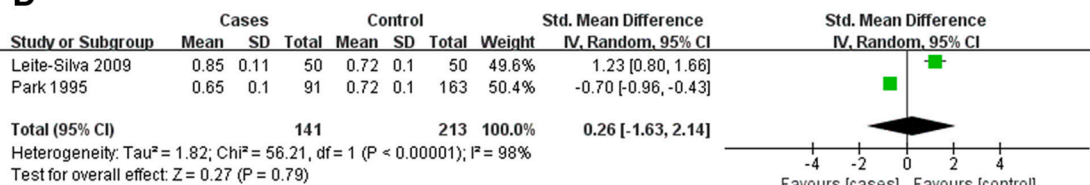
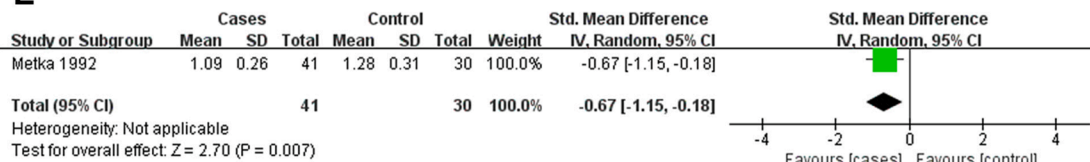
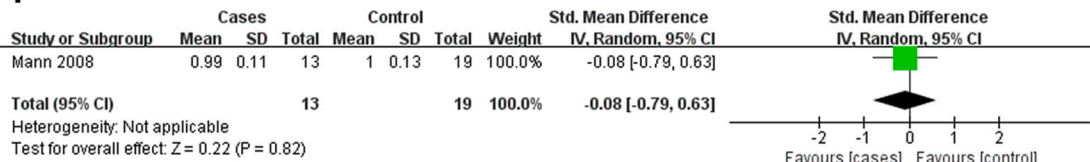
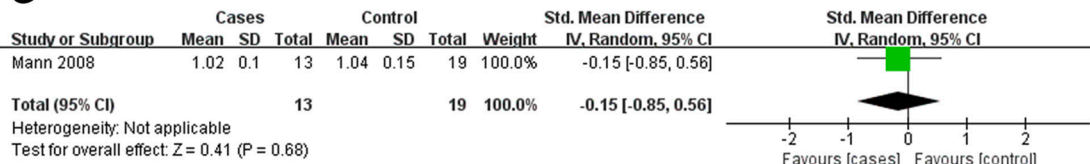
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Fig. 2. Forest plot of the levels of BMD content in cases and healthy controls. Weights are from random effects analysis. (A) Meta-analysis of BMD of lumbar spine. (B) Meta-analysis of BMD of femur neck. (C) Meta-analysis of BMD of Ward's triangle. (D) Meta-analysis of BMD of trochanter. (E) Meta-analysis of BMD of non-dominating forearm. (F) Meta-analysis of BMD of total hip. (G) Meta-analysis of BMD of hipneck. CI, confidence interval; SD, standard difference.

Table 2
Subgroup analysis to investigate the relationship between geographical location and BMD content of lumbar spine, femur neck.

BMD content	Geographical location			
	Europe	Asia	South America	Central Asia
lumbar spine				
No. of studies	4	2	1	2
I ²	16 %	86 %	–	28 %
SMD(95 %CI)	–0.39 (–0.64, –0.14)	–1.15 (–2.58, 0.29)	1.14 (0.72, 1.57)	–0.41(–0.81, 0.01)
P	0.002	0.12	<0.00001	0.04
femur neck				
No. of studies	0	1	1	2
I ²	–	–	–	46 %
SMD(95 %CI)	–	–0.98(–1.25, –0.71)	–1.17 (–1.60, –0.75)	–0.37 (–0.84, 0.10)
P	–	<0.00001	<0.00001	0.12

BMD-bone mineral density;SMD-standard mean difference.

There was significant heterogeneity among the studies including in the meta-analysis of the content of BMD, to search for the sources of heterogeneity and more accurately assess the differences between POI patients and healthy controls, subgroup analyses were conducted by geographical location, number of cases, mean BMI value of cases, mean age of cases, duration of amenorrhea of POI and drug usage.

As for geographical location (Table 2), a lower BMD content of lumbar spine was found in European ($p = 0.002$) and Central Asian ($p = 0.04$) women with POI than in healthy controls, but not in Asian ($p = 0.12$). While a higher BMD content of lumbar spine was found in South American ($p < 0.00001$), and the heterogeneity of Europe ($I^2 = 16\%$) and Central Asia ($I^2 = 28\%$) subgroup decreased. A lower BMD content of femur neck was found in Asian ($p < 0.00001$) and South American ($p < 0.00001$) women with POI than in healthy controls, but not in Central Asian cases ($p = 0.12$), and the heterogeneity of Central Asia subgroup decreased ($I^2 = 46\%$).

As for number of cases (Table 3), in the subgroup with cases < 50 ($p = 0.04$), the BMD content of lumbar spine in women with POI was lower than that in healthy controls, but not in the subgroup with cases ≥ 50 ($p = 0.81$). And the heterogeneity was reduced in the subgroup with the number of cases < 50 ($I^2 = 67\%$). Women with POI had a lower BMD content of femur neck than the healthy controls with cases ≥ 50 ($p < 0.00001$), but not in the subgroup with cases < 50 ($p = 0.12$), the heterogeneity was reduced in this subgroup ($I^2 = 46\%$), and there was no

heterogeneity in the subgroup with the number of cases ≥ 50 ($I^2 = 0\%$).

As for the mean BMI value of cases (Table 3), patients with the mean BMI not available had a lower BMD content of lumbar spine than the healthy controls ($p < 0.0001$), but not in the subgroup with the mean BMI < 24 kg/m² ($p = 0.42$) and the mean BMI ≥ 24 kg/m² ($p = 0.30$), and no heterogeneity was found in the subgroup with the mean BMI not available and the mean BMI ≥ 24 kg/m² ($I^2 = 0\%$). The results showed that patients with the mean BMI < 24 kg/m² ($p < 0.00001$) and the mean BMI not available ($p = 0.03$) had a lower BMD content of femur neck than healthy controls, but not in the subgroup with the mean BMI ≥ 24 kg/m² ($p = 0.36$), and no heterogeneity was found in the subgroup with the mean BMI < 24 kg/m² ($I^2 = 0\%$).

As for the mean age of cases (Table 4), women over the age of 30 with POI had a lower BMD content of lumbar spine than healthy controls ($p = 0.02$), but in the subgroup with the mean age of cases < 30, BMD content of lumbar spine was not associated with POI ($p = 0.83$), and the heterogeneity was reduced in the subgroup with the mean age of cases ≥ 30 ($I^2 = 65\%$). Women under the age of 30 with POI had a lower BMD content of femur neck than healthy controls ($p < 0.0001$), but not in the subgroup with the mean age of cases ≥ 30 ($p = 0.14$), and the heterogeneity was reduced in the subgroup with the mean age of cases < 30 ($I^2 = 44\%$).

As for duration of amenorrhea of POI (Table 4), in the subgroup with the duration < 3 years, there was no correlation of BMD content of lumbar spine in women with POI and healthy controls ($p = 0.29$), and the same conclusion was found in the subgroup with the duration ≥ 3 years ($p = 0.70$), while patients with the duration not available had a lower BMD content of lumbar spine than the healthy controls ($p = 0.003$), the heterogeneity of this subgroup decreased ($I^2 = 61\%$). Patients with the duration ≥ 3 years had a lower BMD content of femur neck than the healthy controls ($p < 0.00001$), but not in the subgroup with the duration not available ($p = 0.12$). Meanwhile, no heterogeneity was found in the subgroup with the duration ≥ 3 years ($I^2 = 0\%$), the heterogeneity of the subgroup with the duration not available decreased ($I^2 = 46\%$).

As for hormone replacement therapy (HRT) usage (Table 5), in the subgroup with HRT not used, the BMD content of lumbar spine in women with POI was lower than that in healthy controls ($p = 0.02$), but not in the subgroup with HRT used ($p = 0.83$), and the heterogeneity was reduced in the subgroup with HRT not used ($I^2 = 64\%$). In the subgroup with HRT not used ($p < 0.0001$) and HRT used ($p = 0.04$), the BMD content of femur neck in the cases was lower than in healthy controls,

As for bone metabolism agents usage (Table 5), in the subgroup with bone metabolism agents used ($p = 0.18$), not used ($p = 0.29$), and not available ($p = 0.08$), the BMD content of lumbar spine was not associated with POI. There was no heterogeneity in the subgroup with bone metabolism agents used not available ($I^2 = 0\%$). Patients with bone metabolism agents not used had a lower BMD content of femur neck

Table 3
Subgroup analysis to investigate the relationship between number of cases,mean BMI value of cases and BMD content of lumbar spine, femur neck.

BMD content	Cases		BMI (kg/m ²)		
	<50	≥ 50	<24	≥ 24	NA
lumbar spine					
No. of studies	5	4	5	2	2
I ²	67 %	94 %	93 %	0 %	0 %
SMD(95 %CI)	–0.49(–0.96, –0.03)	–0.09(–0.82, 0.64)	–0.31(–1.08, 0.45)	–0.18 (–0.52, 0.16)	–0.62(–0.90, –0.33)
P	0.04	0.81	0.42	0.30	<0.0001
femur neck					
No. of studies	2	2	2	1	1
I ²	46 %	0 %	0 %	–	–
SMD(95 %CI)	–0.37(–0.84, 0.10)	–1.04 (–1.27, –0.81)	–1.04 (–1.27, –0.81)	–0.18(–0.57, 0.21)	–0.67(–1.27, –0.08)
P	0.12	<0.00001	<0.00001	0.36	0.03

BMI-body mass index;BMD-bone mineral density;SMD-standard mean difference; NA-not available.

Table 4
Subgroup analysis to investigate the relationship between mean age of cases,duration of amenorrhea of POIand BMD content of lumbar spine, femur neck.

BMD content	Age (years)		Duration (years)		
	<30	≥30	<3	≥3	NA
lumbar spine					
No. of studies	4	5	1	2	6
I ²	94 %	65 %	–	98 %	61 %
SMD(95 %CI)	–0.09(–0.96, 0.77)	–0.46(–0.83, –0.09)	–0.26(–0.75, 0.22)	0.32(–1.29, 1.92)	–0.52(–0.86, –0.17)
P	0.83	0.02	0.29	0.70	0.003
femur neck					
No. of studies	2	2	0	2	2
I ²	44 %	91 %	–	0 %	46 %
SMD(95 %CI)	–0.97(–1.45, –0.49)	–0.59(–1.38, 0.19)	–	–1.04(–1.27, –0.81)	–0.37(–0.84, 0.10)
P	<0.0001	0.14	–	<0.00001	0.12

BMD-bone mineral density;SMD-standard mean difference; NA-not available.

Table 5
Subgroup analysis to investigate the relationship between drug usage and BMD content of lumbar spine, femur neck.

BMD content	HRT usage		Bone metabolism agents usage		
	<30	≥30	Yes	No	NA
lumbar spine					
No. of studies	4	5	1	6	2
I ²	94 %	65 %	–	92 %	0 %
SMD(95 %CI)	–0.09(–0.96, 0.77)	–0.46(–0.83, –0.09)	–0.26(–0.65, 0.12)	–0.35(–1.01, 0.30)	–0.32(–0.67, 0.04)
P	0.83	0.02	0.18	0.29	0.08
femur neck					
No. of studies	2	2	1	3	0
I ²	44 %	91 %	–	0 %	–
SMD(95 %CI)	–0.97(–1.45, –0.49)	–0.59(–1.38, 0.19)	–0.18(–0.57, 0.21)	–0.99(–1.20, –0.78)	–
P	<0.0001	0.14	0.36	<0.00001	–

BMD-bone mineral density;SMD-standard mean difference; NA-not available.

than the healthy controls ($p < 0.00001$), but not in the subgroup with the bone metabolism agents used ($p = 0.36$), and no heterogeneity was found in the subgroup with the bone metabolism agents not used ($I^2 = 0\%$).

Sensitivity analysis

The studies were removed one by one to check the stability and reliability of the meta-analysis results. As for the BMD content of lumbar spine, there was no qualitative change in the total effect size after removing the studies one by one, indicating that the meta-analysis results were stable and reliable [24]. The heterogeneity decreased ($I^2 = 48\%$; $p = 0.06$), and the BMD content of lumbar spine in women with POI changed to been lower than that in healthy controls (SMD: -0.47 ; 95 % CI: -0.70 to -0.24 ; $P < 0.0001$). For BMD content of femur neck, the results of this meta-analysis showed weak stability and sensibility compared with Uygur’s findings [22]. In the absence of this study, the heterogeneity decreased ($I^2 = 0\%$; $p = 0.41$), and the BMD content of femur neck in women with POI remained lower than in controls (SMD: -0.99 ; 95 % CI: -1.20 to -0.78 ; $P < 0.00001$).

Publication bias

As the Cochrane Handbook for Systematic Reviews of Interventions (<https://www.cochranehandbook.org>) stated that multiple studies not gathered with 10 studies yield unreliable results, publication bias was not assessed.

Discussion

Primary ovarian insufficiency, due to the long-term complications

such as cardiovascular disease and osteoporosis, has been regarded as “serious chronic disease with far reaching effects on physical and emotional health” [26]. The measurement of bone metabolism indicators combined with BMD can effectively reflect the process of bone turnover and predict the risk of fracture in patients with osteoporosis. 25-hydroxyvitamin D3 (25[OH]VD3) can stimulate the activity of osteoblasts, promote bone calcium deposition and bone formation, improve the activity of osteoclasts, enhance the effect of osteocalcin on bone, accelerate the reabsorption of calcium and phosphorus in renal tubules, and reduce the excretion of calcium and phosphorus in urine [27]. Osteocalcin which is a non-collagenous bone matrix protein synthesized and secreted by osteoblasts reflect the activity of osteoblasts and bone metabolic level, and play an important role in bone development [28]. Research shows that bone mineral density and 25(OH)VD3 contents in patients with POI are lower than those in healthy controls, while the level of osteocalcin is higher than that of healthy women of the same age [29], which results in agreement with those of this meta-analysis. Therefore, premature menopause has adverse long-term effects on bone health.

The benefits of intact ovarian function for bone health can be generally attributed to estradiol. Estradiol has a positive regulatory effect on promoting bone growth [30]. It has also proven that endogenous estradiol inhibits the effect of osteoblasts which produces inflammatory factors such as interleukin and tumor necrosis factor, increases the content of osteoprotegerin, calcitonin, vitamin D and transforming growth factor, promotes proliferation and differentiation of osteoblasts [31]. Estradiol can inhibit the binding of receptor activator of NF-KappaB and osteoclasts precursors, lead to apoptosis of osteoclasts and inhibit osteoclasts activity, and shorten the survival time of osteoclasts [32]. Meanwhile, due to the activation of the ER/AKT signaling pathway, estradiol deficiency leads to an increase in bone fragility,

which seriously affects the quality of life of patients [33]. Estradiol deficiency increases the decomposition of phosphate and bone trabecular matrix components, reduces the reabsorption of calcium ions, phosphorus factors, etc., and promotes bone loss [34]. Decreased estradiol reduces bone sensitivity to mechanical stimulation, leads to pathological changes similar to disuse bone loss [35]. In addition, estradiol deficiency increases the level of oxidative stress, which leads to inhibition of osteogenesis and activation of osteoclast [36,37]. A gradual decrease in BMD occurs in patients with premature ovarian failure as estradiol deficiency continues, and osteoporosis develops earlier [38]. Therefore, women with POI are more likely to suffer fracture morbidity and mortality later in life as a result of estradiol deficiency for a longer period of time than women with natural menopause.

In recent years, several international guidelines recommended hormone replacement therapy (HRT) for women with POI [9]. There has been substantial research undertaken on the role of exogenous supplemental hormone. The benefits of HRT are many, including relief of clinical symptoms, prevention of occurrence and development of POI, reduction of high disability rates and death rates, reduction of long-term complications, as well as improving living conditions of patients with POI [9]. It has been reported that HRT can increase the BMD content [39]. Study has shown that the structure of the trabecular bone is more complete after 2 years of HRT [40]. In 2016, the International Society of Menopause noted that HRT should be seen as a first-line treatment for fracture prevention [41]. In 2017, the North American Menopause Society noted that 6 fractures of the body were reduced per 10,000 person-years due to the use of HRT [42]. Compared with postmenopausal women who have never used HRT, women who have used HRT for more than 7 years have a reduced risk of distal forearm and proximal humerus fracture [43]. Moreover, HRT protects the spine by virtue of maintain the size of the individual disc the total disc space and the length of the spine, contrary to bisphosphonates [44]. From the perspective of health economics, meanwhile, HRT is more cost-effective than drugs such as bisphosphate and calcitonin, and has systemic benefits other than anti-osteoporosis effects [45]. But in reality, there is an extremely low acceptance rate for HRT among women with POI [46].

Subgroup analysis was conducted to further explore the source of heterogeneity and more accurately evaluate the correlation between BMD content and POI. Heterogeneity in this *meta*-analysis of the correlation between BMD content of lumbar spine, femur neck and POI might come from region, number of cases, mean BMI value of cases, mean age of cases, duration of amenorrhea of POI, HRT use, as well as bone metabolism agents use. Meanwhile, due to the use of HRT, the conclusion on the BMD content of Ward's triangle and trochanter in Leite-Silva's study was contrary to Park's study. (1) People in different regions have different dietary habits, lifestyles and economic levels, which have different effects on BMD content; (2) most of the references included in this *meta*-analysis were designed as case-control studies. In case-control study, if the number of cases is enormous, the research results will be affected by more confounders; (3) Elevated BMI has a protective effect on the prevention of osteoporosis [47]. Therefore, BMI value might be one of the sources of heterogeneity; (4) ovarian reserve function is related to physiological age, when age is over 35 years, ovarian function significantly declines [48]. Studies have shown that BMD increases with age before 30 years, while after peak bone mass, the prevalence of osteoporosis shows an increasing trend with physiological age [49]. Thus physiological age was regarded as one of the sources of heterogeneity; (5) the longer menopause lasts, thus the longer estrogen deficiency lasts. Therefore, duration of amenorrhea of POI might be one of the sources of heterogeneity; and (6) both HRT and bone metabolism agents (e.g., calcium supplementation) can improve BMD. Hence HRT use or bone metabolism agents use might be one of the sources of heterogeneity.

Despite finding some sources of heterogeneity by subgroup analysis, the heterogeneity was still evident, so a sensitivity analysis was performed. In this *meta*-analysis, the combined results of the BMD content of

lumbar spine and POI were sensitive to Leite-Silva's study [24]. After excluding the reference, the heterogeneity decreased (I^2 decreased from 87 % to 48 %), and P value of the combined result was less than 0.05 (P value changed from 0.14 to < 0.0001). Therefore, Leite-Silva's study [24] had significant impact on the heterogeneity of the combined results. After careful reading of the reference, we found thirty-five of the patients were HRT users with a mean duration of 28 months of use in this reference. Hence, HRT use might be the main source of heterogeneity in this reference. In the meantime, the combined results of the BMD content of femur neck and POI were sensitive to Uygun's study [22], and the P values of the combined results were all less than 0.05 before and after the exclusion of the reference. However, after excluding the reference, the heterogeneity decreased (I^2 decreased from 80 % to 0 %), therefore, Uygun's study [22] had significant impact on the heterogeneity of the combined results. After careful reading of the reference, we found intermittent use of HRT in women with POI, and thirty-six of the patients received calcium supplementation in this reference. Therefore, HRT and bone metabolism agents use might be the main sources of heterogeneity in this reference.

Nonetheless, there are however a number of limitations to this *meta*-analysis should be acknowledged. Obvious heterogeneity existed among the BMD content of original studies due to differences in sample size, background of subjects, and the detection reagent used to detect the content of BMD. There was no association between POI and the BMD content of lumbar spine, Ward's triangle, trochanter, as well as hip due to the effect of HRT and bone metabolism agents. In addition, insufficient number of references was also one of the reasons. Therefore, more references are needed to verify this result. Furthermore, it is unclear whether the main confounding factors (e.g., dietary habit and physical activity) affecting the BMD content of patients in original studies were adjusted for. Lastly, the references in this study were mostly case-control studies, and hence limiting causal inference. Therefore, more cohort studies are needed to predict how BMD content changes over time. The references included in this study were almost case-control studies, which would limit causal inference. In these circumstances, we therefore recommend a conservative interpretation of our conclusions.

Conclusion

This review showed that the BMD content in femur neck and non-dominating forearm were significantly lower in subjects with POI. In a way, women whose ovarian function is lost at a very early age will likely experience bone mass loss, leading to a higher risk of fractures. Based on this result, we strongly recommend that patients with POI should undergo comprehensive BMD evaluations in order to assure that early medical intervention (especially HRT) works as intended and minimize fracture morbidity and mortality associated with osteopenia.

Author contributions

Mei Jiang: Conceptualization, Formal analysis, Methodology, Writing – original draft, Writing – review and editing. Ying Gao: Formal analysis, Methodology, Writing – original draft. Hongping Hou: Investigation, Software, Writing – review and editing. Jing Guo: Visualization, Writing – review and editing. Wenyuan Li: Supervision, Writing – review and editing. Tianyu Qin: Supervision, Writing – review and editing. Minglu Shi: Supervision, Writing – review and editing. Ling Huang: Conceptualization, Formal analysis, Resources, Supervision, Writing – review and editing. All authors reviewed and approved the final version and no other person made a substantial contribution to the paper.

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CRediT authorship contribution statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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